

HCM360 HRIS

Deployment Setup

Run, deploy, and operate HCM360 HRIS v1.0

Version 1.0.0 · April 2026

1. System Requirements

HCM360 is delivered as a Docker Compose stack. All dependencies run in containers — the host only needs Docker Engine.

Component	Minimum	Recommended
CPU	2 cores	4+ cores
RAM	4 GB	8+ GB
Disk	20 GB	50+ GB (for uploads + backups)
Docker Engine	20.10+	24+ with Compose v2
OS	Linux / macOS / Windows WSL2	Linux (Ubuntu 22.04 LTS)

2. Repository Layout

```
hcm360/
■■■ app.py                # Flask entry point
■■■ config.py             # Feature flags from env
■■■ requirements.txt       # pip deps
■■■ Dockerfile            # Web container build
■■■ docker-compose.yml    # 3-service orchestration
■■■ docker/init_db.sh     # First-run DB initializer
■■■ .env                  # API keys (not committed)
■■■ db/                   # Versioned SQL migrations (00-51)
■■■ modules/              # Blueprints per domain
■■■ services/             # Shared services
■■■ templates/            # Jinja2 templates
■■■ uploads/              # File uploads (bind-mounted)
■■■ backups/              # pg_dump files
■■■ scripts/restore_v1.sh # Restore helper
■■■ payroll/              # Standalone payroll microservice
■■■ VERSION               # 1.0.0
■■■ CHANGELOG.md          # Release notes
```

3. First-Time Setup

3.1 Clone and Configure

- 1 Clone the repository to /opt/hcm360 (or any path)
- 2 Copy .env.example to .env and set ANTHROPIC_API_KEY for ARIA AI (optional)
- 3 Review docker-compose.yml — port mappings default to 8093 (web), 8095 (payroll), 5440 (DB)
- 4 Build and start: docker compose up -d --build
- 5 Watch initial DB seed: docker compose logs -f hris_db
- 6 Once hris_db is healthy, web starts automatically
- 7 Open <http://localhost:8093> and pick a demo user

3.2 First-Run DB Initialization

The db/ folder is bind-mounted into /docker-entrypoint-initdb.d/sql on the PostgreSQL container. docker/init_db.sh runs every .sql file in numeric order (00_extensions through 51_workforce_planning_access). This happens only on first start when the data volume is empty.

4. docker-compose.yml

Three services share a default network. The DB uses a named volume so data survives docker compose down and rebuilds.

```
services:
  hris_db:
    image: postgres:15
    restart: unless-stopped
    environment:
      POSTGRES_DB: hris_db
      POSTGRES_USER: hris_admin
      POSTGRES_PASSWORD: hris_secure_pw
    ports: ["5440:5432"]
    volumes:
      - hris_db_data:/var/lib/postgresql/data
      - hris_uploads:/uploads
      - ./db:/docker-entrypoint-initdb.d/sql:ro
      - ./docker/init_db.sh:/docker-entrypoint-initdb.d/init_db.sh:ro
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U hris_admin -d hris_db"]

  hris_web:
    build: .
    image: hcm360-hris_web:v1.0
    restart: unless-stopped
    depends_on:
      hris_db: {condition: service_healthy}
    environment:
      DATABASE_URL: postgresql://hris_admin:hris_secure_pw@hris_db:5432/hris_db
    volumes:
      - ./app
      - hris_uploads:/uploads
    ports: ["8093:5000"]

  payroll_web:
    build: ./payroll
    image: hcm360-payroll_web:v1.0
    restart: unless-stopped
    depends_on:
      hris_db: {condition: service_healthy}
    ports: ["8095:5001"]

volumes:
  hris_db_data:
    name: hcm360_hris_db_data
  hris_uploads:
    name: hcm360_hris_uploads
```

5. Environment Variables

Variable	Default	Purpose
DATABASE_URL	postgresql://hris_admin:...@hris_db:5432/hris_db	PostgreSQL connection URL
SECRET_KEY	changeme-hris-prod	Flask session signing key — CHANGE in production
UPLOAD_FOLDER	/uploads	Bind-mounted folder for document uploads
ANTHROPIC_API_KEY	(empty)	Enables ARIA AI features when set
ENABLE_ATTENDANCE	true	Register attendance blueprint
ENABLE_LEAVE	true	Register leave blueprint
ENABLE_PAYROLL	true	Register payroll blueprint
ENABLE_RSP / PM / LD / RR	true	Register recruitment, performance, learning, rewards
ENABLE_DMS / DISCIPLINE / HEALTH	true	Register document management, discipline, health & safety
ENABLE_ORGCHART	true	Register org chart blueprint
ENABLE_AI / ANALYTICS	true	Register AI and analytics modules

6. Deploying Updates

6.1 Code Changes Only

The web container bind-mounts the project root at /app. Edit files locally and restart the container to pick up Python changes:

```
# Restart web only
docker compose restart hris_web

# Or kill and start fresh (keeps DB data)
docker compose down hris_web && docker compose up -d hris_web
```

6.2 Schema Changes

Add a new numbered SQL file under db/ (e.g., db/52_my_feature.sql). For existing deployments, apply manually since db/ only runs on empty volumes:

```
docker exec -i hris_db psql -U hris_admin -d hris_db < db/52_my_feature.sql
```

6.3 Dockerfile / Dependency Changes

```
docker compose build --no-cache hris_web
docker compose up -d hris_web
```

7. Data Persistence

Two named Docker volumes ensure data survives container rebuilds:

Volume	Mount	Contents
hcm360_hris_db_data	/var/lib/postgresql/data	All PostgreSQL data files
hcm360_hris_uploads	/uploads	Document uploads, logos, certificates

Note: docker compose down by itself does NOT remove volumes. Only docker compose down -v will delete them. Always use docker compose down (without -v) unless you intend a clean wipe.

8. Backups & Restore

8.1 Manual Backup

```
docker exec hris_db pg_dump -U hris_admin -d hris_db --format=custom \
--file=/tmp/hcm360_backup.dump
docker cp hris_db:/tmp/hcm360_backup.dump ./backups/hcm360_$(date +%Y%m%d).dump
```

8.2 Scheduled Backup (cron)

```
# Add to host crontab
0 2 * * * cd /opt/hcm360 && docker exec hris_db pg_dump -U hris_admin \
-d hris_db --format=custom --file=/tmp/backup.dump \
&& docker cp hris_db:/tmp/backup.dump \
/opt/hcm360/backups/hcm360_$(date +%Y%m%d).dump
```

8.3 Restore

```
./scripts/restore_v1.sh backups/hcm360_v1.0_20260412.dump
```

9. Monitoring & Logs

Command	Purpose
docker compose ps	Show container status
docker compose logs -f hris_web	Tail web logs
docker compose logs -f hris_db	Tail DB logs
docker exec hris_db psql -U hris_admin -d hris_db	Interactive psql shell
docker stats	Live CPU/memory per container

10. Production Hardening Checklist

- Change POSTGRES_PASSWORD and SECRET_KEY to strong unique values
- Put the web behind a reverse proxy (nginx / Caddy) with TLS
- Restrict 5440 (DB port) to localhost only — do not expose publicly

- Enable automated off-site backups (daily pg_dump + S3 sync)
- Set up log shipping (Loki / CloudWatch / Datadog)
- Configure SMTP for notification emails
- Review and tighten access matrix at /admin/access-matrix
- Review field privacy rules at /admin/privacy
- Audit user list and disable default demo users (superadmin, hradmin, etc.)
- Replace ANTHROPIC_API_KEY with a production key if AI features are enabled

11. Troubleshooting

Symptom	Likely Cause	Resolution
403 Forbidden on valid URL	Access matrix denies role	Add role to core.role_page_access for the page
Login page loops	Session cookie domain mismatch	Check SECRET_KEY is stable and domain matches
Empty dropdowns	Reference data missing	Run seed files 17_seed_core.sql and 18_seed_modules.sql
PDF download fails	reporting_service missing font	Verify ReportLab install in hris_web container
DB container won't start	Volume corrupted	Restore from latest backup using scripts/restore_v1.sh
Slow pages	Missing index	EXPLAIN ANALYZE the slow query and add index

12. Uninstalling

To completely remove HCM360 including all data:

```
docker compose down -v # stop + remove volumes
docker image rm hcm360-hris_web:v1.0 \
                hcm360-payroll_web:v1.0 \
                postgres:15
rm -rf /opt/hcm360
```